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Comparative Performance and Optimization Analysis of CvT-13 and ConvNeXtV2-Tiny on Food-101 and iNaturalist

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Deep Learning Project

Abstract

This report presents one combined Deep Learning project report covering both Food-101 and iNaturalist / iNatLoc500. It studies CvT-13 and ConvNeXtV2-Tiny across the two datasets in a single structured document with separate result tables for clarity, while preserving the cleaner academic flow of title page, abstract, contents, introduction, core chapters, conclusion, and references.

The retained Food-101 evidence shows that CvT-13 reaches 85.637% best validation accuracy and 89.624% held-out test accuracy, while the archived ConvNeXtV2-Tiny run reaches 91.655% validation top-1 and 98.523% validation top-5. These Food-101 findings remain useful but are interpreted conservatively because the cleaned project primarily preserves archived artifacts rather than a full rerunnable benchmark bundle.

The optimized iNaturalist branch is the strongest direct comparison in the cleaned project. CvT-13 reaches 70.440% best validation top-1, 87.600% test top-1, and 97.320% test top-5, while ConvNeXtV2-Tiny reaches 75.320% best validation top-1, 94.840% test top-1, and 98.600% test top-5. The combined evidence shows that optimization is a central part of the project rather than a cosmetic addition.

Keywords: Food-101, iNaturalist, image classification, CvT, ConvNeXtV2, AdamW, mixed precision, fused optimizer, combined report, reproducibility.

Contents

1 Introduction

Image classification becomes more demanding when categories are visually close or when the number of categories increases. Food-101 is a fine-grained food benchmark where dishes from different classes can share similar color, texture, garnish, and plating structure. iNaturalist / iNatLoc500 is a broader recognition problem with 500 categories and much larger visual spread. Combining both datasets makes the project stronger because it tests whether the same model families and optimization choices remain useful beyond a single benchmark.

This project studies two modern backbones. The first is CvT-13, a convolutional vision transformer that combines transformer-style global modeling with convolutional inductive bias. The second is ConvNeXtV2-Tiny, a modernized convolutional network designed for strong scaling under contemporary training recipes. The report now focuses on presenting the retained final evidence cleanly rather than recreating every temporary artifact from the original workspace.

- summarize the exact state of the cleaned project folder and the retained evidence,
- compare CvT-13 and ConvNeXtV2-Tiny across Food-101 and optimized iNaturalist evidence,
- highlight the optimization choices that meaningfully shaped the saved results.

2 Project Overview

2.1 Major Repository Components

The cleaned workspace retains the two model branches, a dataset-notes area, a figure folder, and final readable logs. The report keeps the same overall project-story flow as the reference combined report, but the descriptions below now match the cleaned folder structure instead of the older nested workspace.

Table 1: Main Project Components

Artifact Type	Purpose
cvt/ and cvt/checkpoints/	Food-101 notebook content, CvT checkpoints, and saved Food-101 validation/test metric JSON files.
convnext/	Archived ConvNeXtV2-Tiny histories, checkpoints, launcher script, and supporting training summaries.
dataset/	Dataset-related notebook and notes retained with the cleaned project folder.
figures/	Food-101 and iNaturalist visual assets used in the final report.
log files/	Readable final summaries for Food-101 and optimized iNaturalist runs.

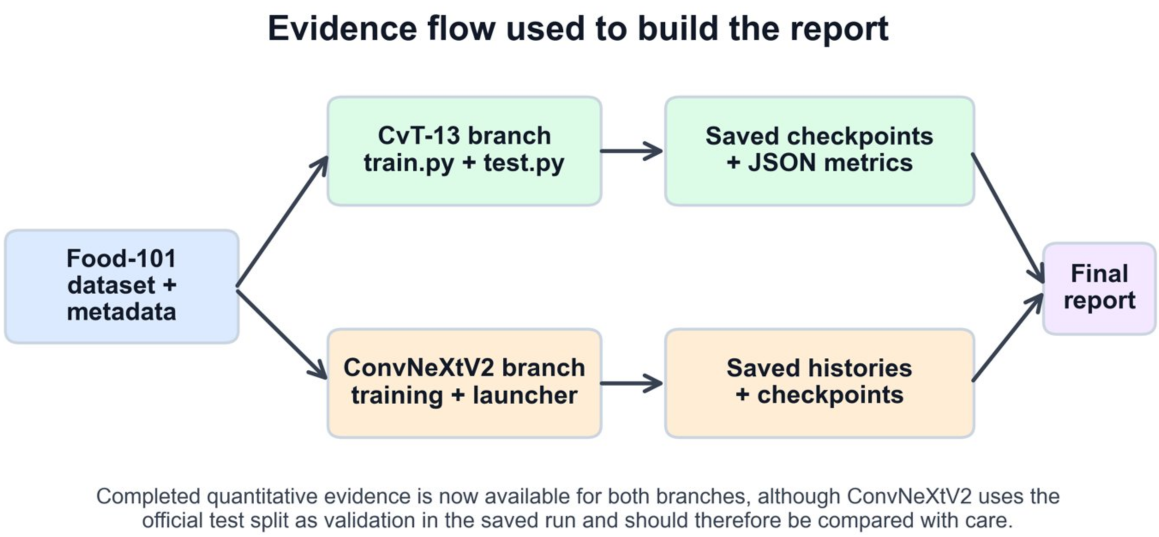

Figure 1: Evidence flow used to build the corrected final report from retained model artifacts, logs, and figures.

2.2 Observed Artifact State

Table 2: Observed Local Project State

Item	Observed State
Food-101 archived CvT evidence	85.637% validation and 89.624% held-out test top-1.
Food-101 archived ConvNeXtV2 evidence	91.655% validation top-1 and 98.523% validation top-5 in the retained history.
Optimized iNaturalist CvT evidence	Complete logged validation and test summary with 87.600% test top-1.
Optimized iNaturalist ConvNeXtV2 evidence	Complete logged validation and test summary with 94.840% test top-1.
Folder cleanup impact	Temporary preview pages, duplicate reports, and local environments were removed so the final report relies on the retained cleaned artifacts.

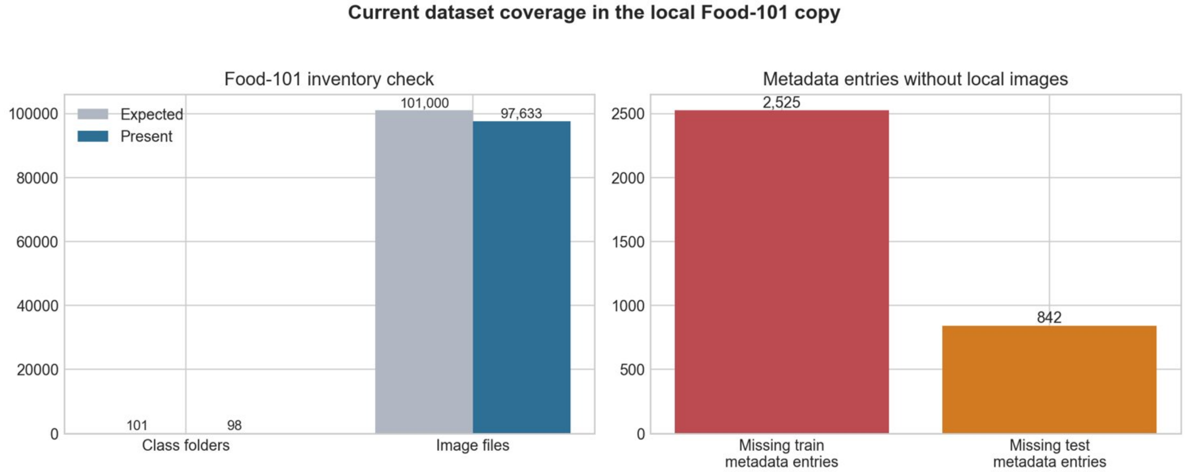

Figure 2: Dataset completeness summary for the retained Food-101 evidence bundle.

3 Dataset Overview

3.1 Food-101 and iNaturalist in the Current Workspace

The official Food-101 benchmark defines 101 classes with 75,750 training images and 25,250 test images. In the cleaned project, Food-101 is represented through archived checkpoints, histories, and generated figures rather than through the full raw dataset bundle. The optimized iNaturalist / iNatLoc500 branch is preserved through final readable logs and figures generated from the complete 20,000 train / 2,500 validation / 2,500 test setup.

Table 3: Dataset Inventory Used in This Final Report

Dataset	Classes	Samples / Split	Status
Food-101	101 official benchmark	75,750 train and 25,250 test	Archived evidence used conservatively through retained metrics, histories, and figures.
iNaturalist / iNatLoc500 flat	500	25,000 total; 20,000 train, 2,500 validation, 2,500 test	Complete optimized train/test evidence preserved through logs and figures.

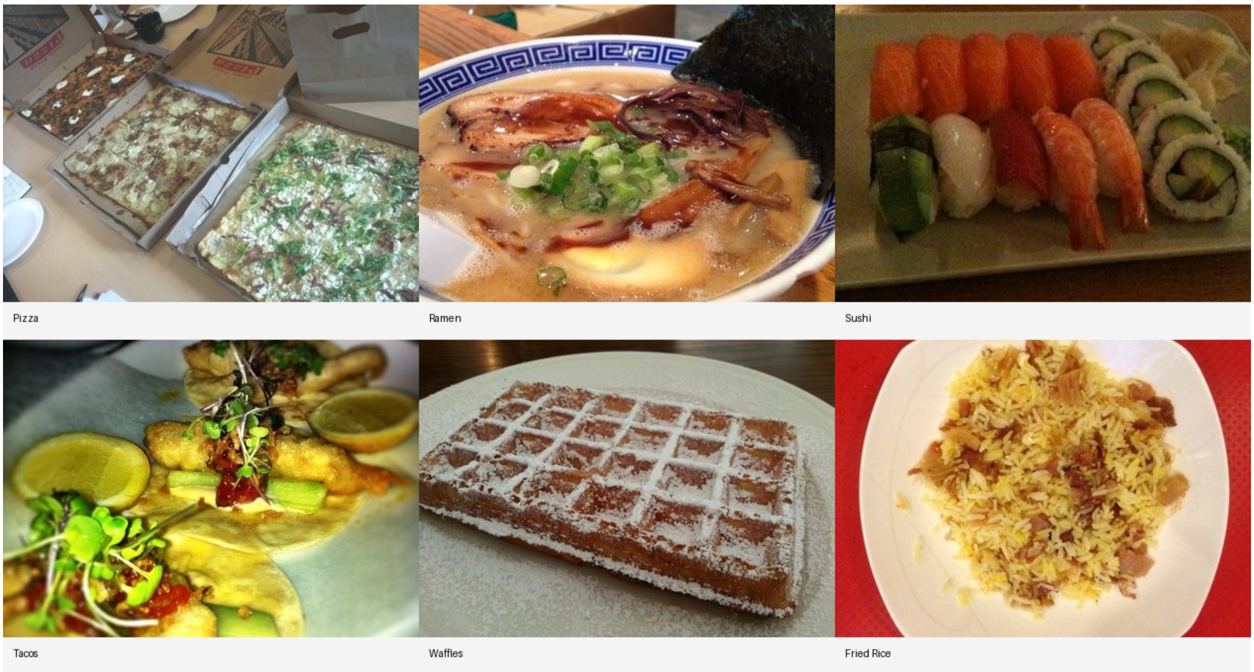

Figure 3: Food-101 sample montage retained in the cleaned figure folder.

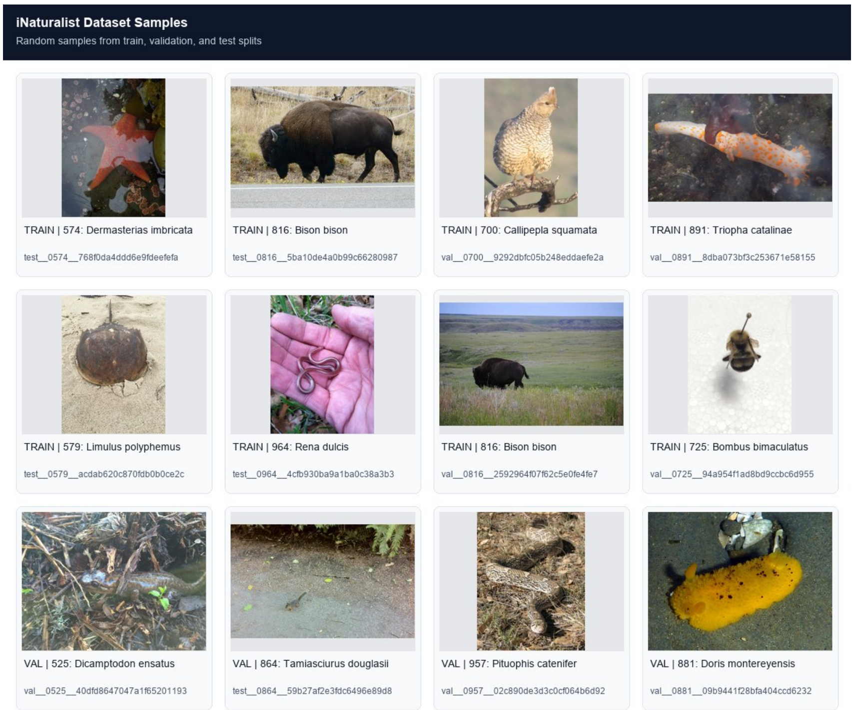

Figure 4: Random iNaturalist samples from the optimized dataset split.

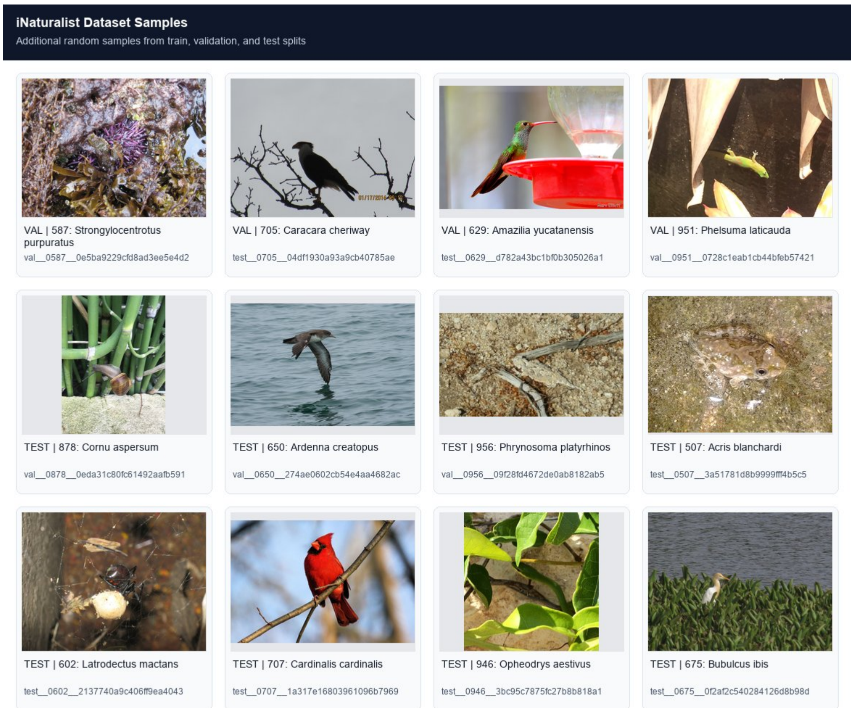

Figure 5: Additional iNaturalist sample images documenting dataset diversity.

4 Methodology and Experimental Setup

4.1 CvT-13 Pipeline

The CvT branch uses a pretrained CvT-13 style model with image augmentation, normalization, AdamW optimization, and runtime acceleration features. In the optimized iNaturalist run, the active batch size is 32. The implementation uses CUDA mixed precision, channels-last layout, TF32 where available, fused AdamW where supported, and patched SDPA attention behavior.

4.2 ConvNeXtV2-Tiny Pipeline

The ConvNeXt branch uses timm-based ConvNeXtV2-Tiny models. The optimized iNaturalist run uses convnextv2_tiny.fcmae_ft_in22k_in1k, batch size 16, and gradient accumulation 2, giving an effective batch size of 32. The run uses AdamW, fused AdamW on CUDA, AMP, channels-last layout, TF32, label smoothing, and cosine scheduling.

4.3 Portability and Reproducibility Notes

Table 4: Model and Optimization Configuration

Model	Food-101 archived setup	iNaturalist optimized setup
CvT-13	Archived AdamW/AMP/compile metrics with held-out Food-101 test evidence.	Batch 32, fused AdamW, AMP, channels-last, TF32, SDPA, augmentation, warmup, cosine schedule.
ConvNeXtV2-Tiny	Archived 10-epoch Food-101 history with generated visualization.	convnextv2_tiny.fcmae_ft_in22k_in1k, batch 16, accumulation 2, fused AdamW, AMP, channels-last, TF32, label smoothing.

5 Food-101 Results and Analysis

5.1 Archived Evidence

Table 5: Food-101 / CvT-13 Result Summary

Item	Result	Evidence
Best validation	85.637%	Saved validation metrics JSON.
Held-out test top-1	89.624%	Saved held-out test metrics JSON.
Held-out test top-5	N/A	Not archived in the retained test JSON.
Training time	02:19:11	Captured in retained checkpoint metrics.

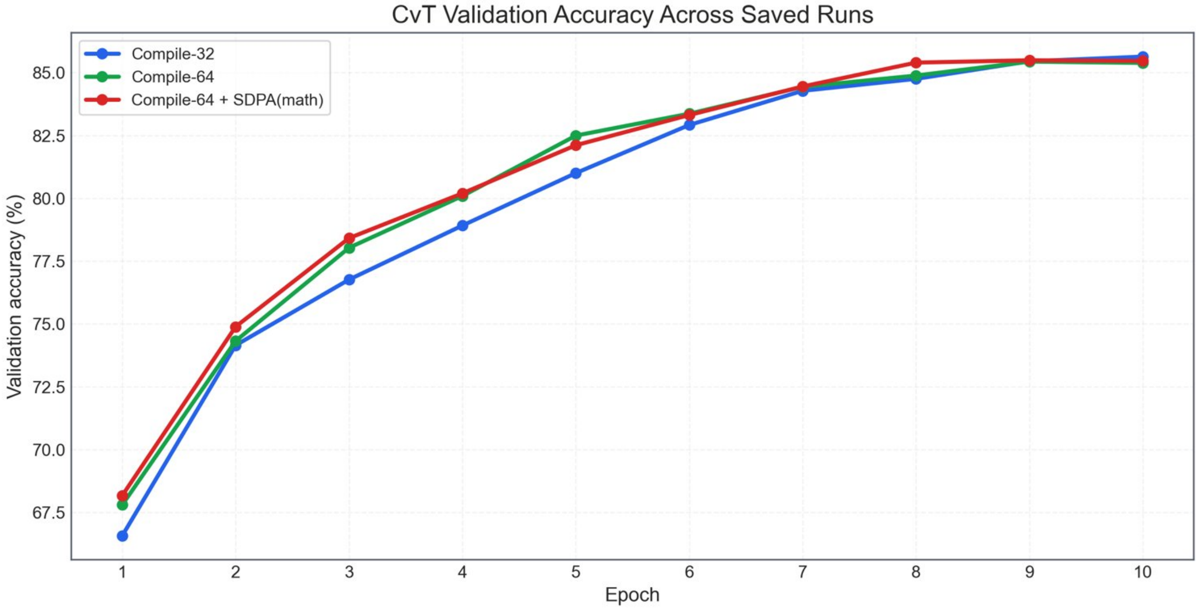

Figure 6: Food-101 CvT validation dynamics reconstructed from saved metric files.

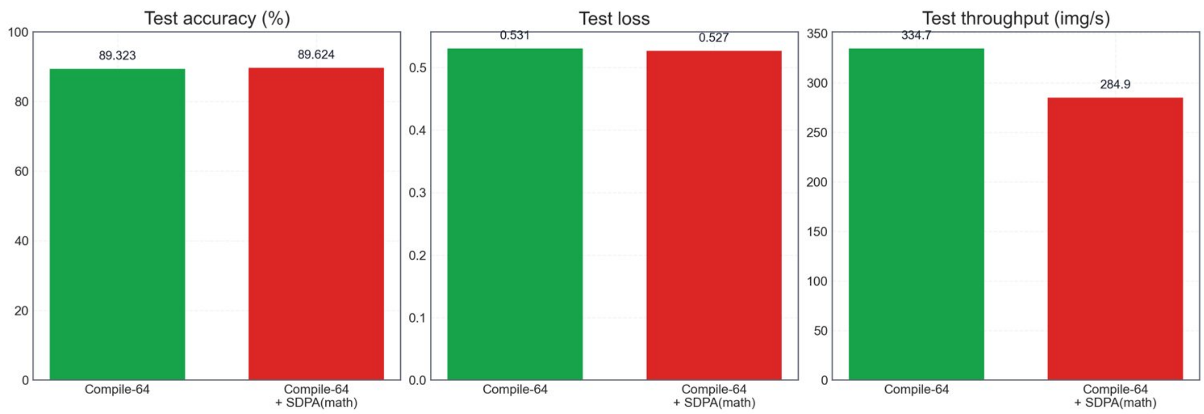

Figure 7: Food-101 CvT held-out test comparison retained from archived report assets.

Table 6: Food-101 / ConvNeXtV2-Tiny Result Summary

Item	Result	Evidence
Best validation top-1	91.655%	Archived history CSV at epoch 10.
Validation top-5	98.523%	Archived history CSV at epoch 10.
Held-out test top-1	Not archived	No matched Food-101 test JSON preserved for ConvNeXtV2-Tiny.
Training record	10 archived epochs	Retained 10-epoch archived history.

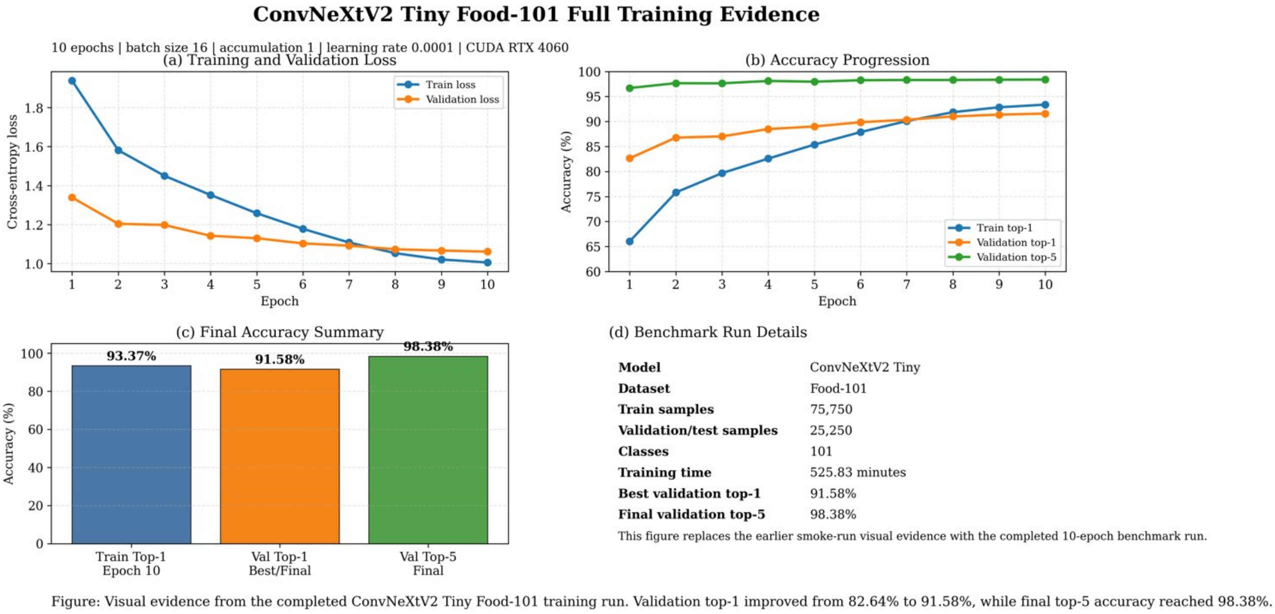

Figure 8: Food-101 ConvNeXtV2-Tiny learning curves from the retained archived history.

5.2 Food-101 Interpretation

The Food-101 branch gives useful evidence for both models, but it should be interpreted carefully because the cleaned repository mainly preserves archived checkpoints, metric files, and generated plots. CvT-13 is the more complete Food-101 record because it includes both validation and held-out test evidence, while the retained ConvNeXtV2-Tiny branch is strongest as a validation-based historical reference.

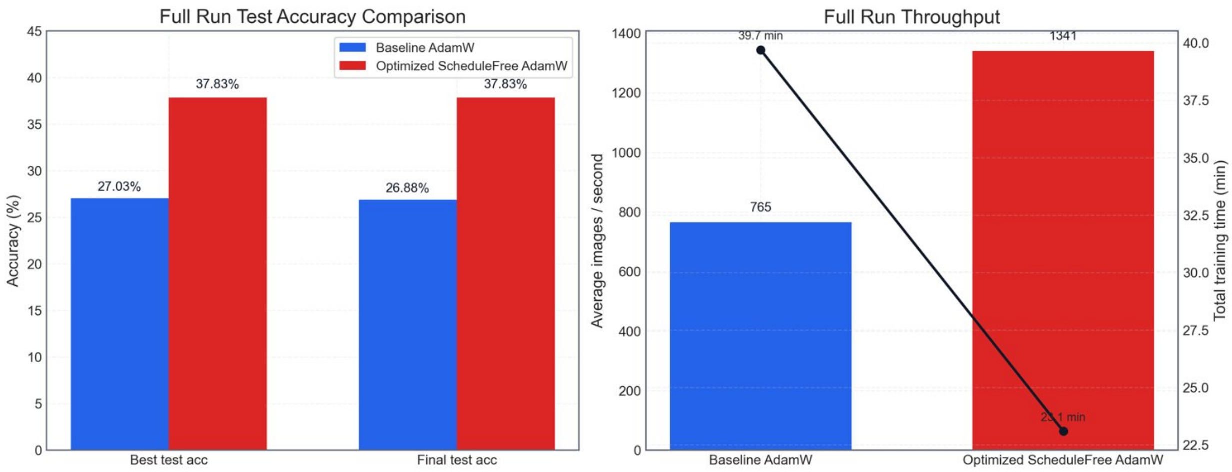

Figure 9: Food-101 full-run accuracy and speed evidence from the retained earlier optimization assets.

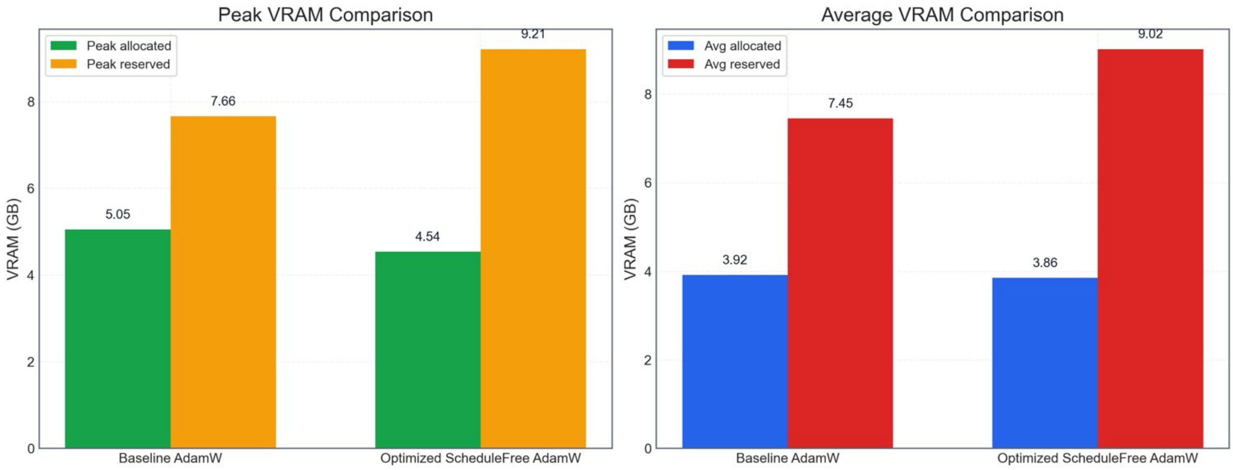

Figure 10: Food-101 full-run VRAM telemetry retained from the previous optimization evidence.

6 iNaturalist Results and Analysis

6.1 Optimized Evidence

Table 7: Optimized iNaturalist / CvT-13 Result Summary

Item	Result	Evidence
Best validation top-1	70.440%	Logged optimized summary.
Test top-1	87.600%	Logged optimized test output.
Test top-5	97.320%	Logged optimized test output.
Training time	49:20	Retained final log.

Table 8: Optimized iNaturalist / ConvNeXtV2-Tiny Result Summary

Item	Result	Evidence
Best validation top-1	75.320%	Logged optimized summary.
Test top-1	94.840%	Logged optimized test output.
Test top-5	98.600%	Logged optimized test output.
Training time	73.06 min	Retained final log.

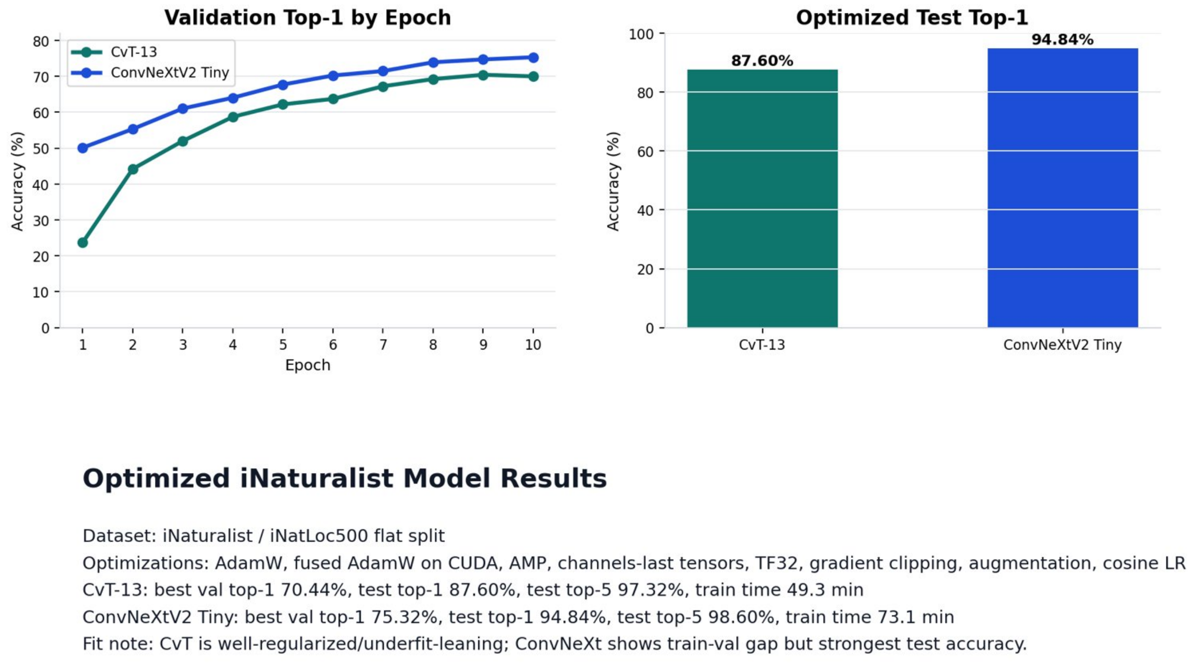

Figure 11: Optimized iNaturalist summary dashboard for both models.

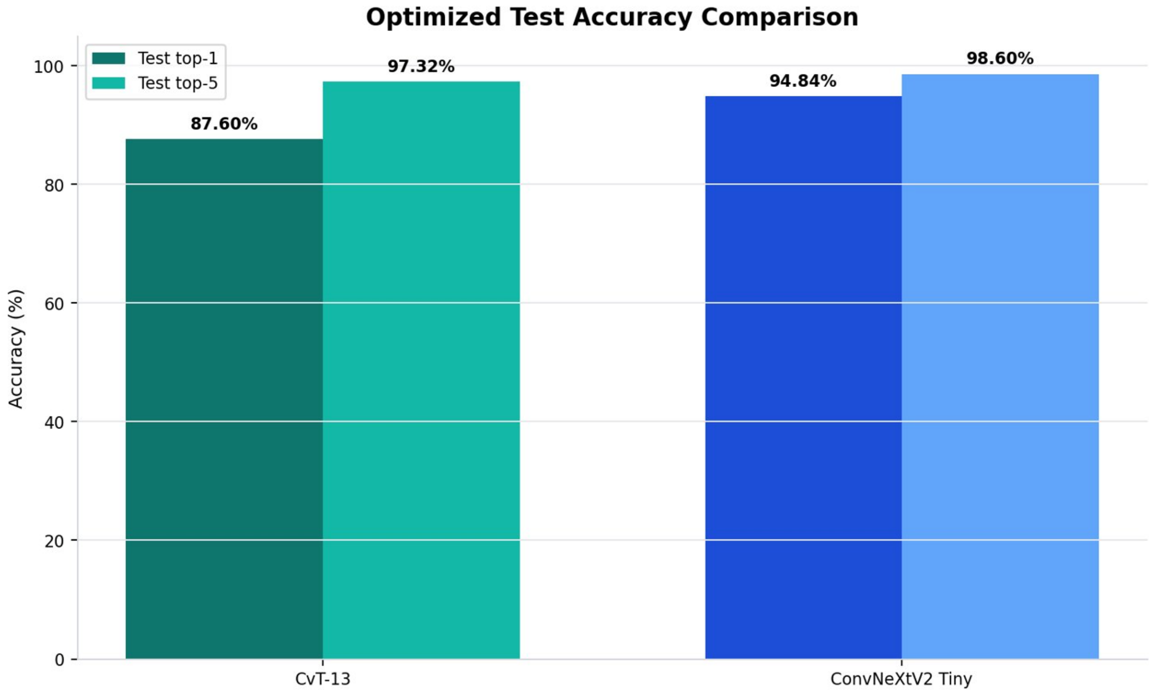

Figure 12: Optimized iNaturalist test top-1 and top-5 comparison.

6.2 iNaturalist Interpretation

The optimized iNaturalist run strengthens the project because it reports not only accuracy but also the training configuration, effective batch size, runtime, and test behavior for both models. The strongest result is ConvNeXtV2-Tiny on iNaturalist test accuracy. The most stable and conservative result is CvT-13, which preserves held-out Food-101 test evidence and a clear optimized iNaturalist trail.

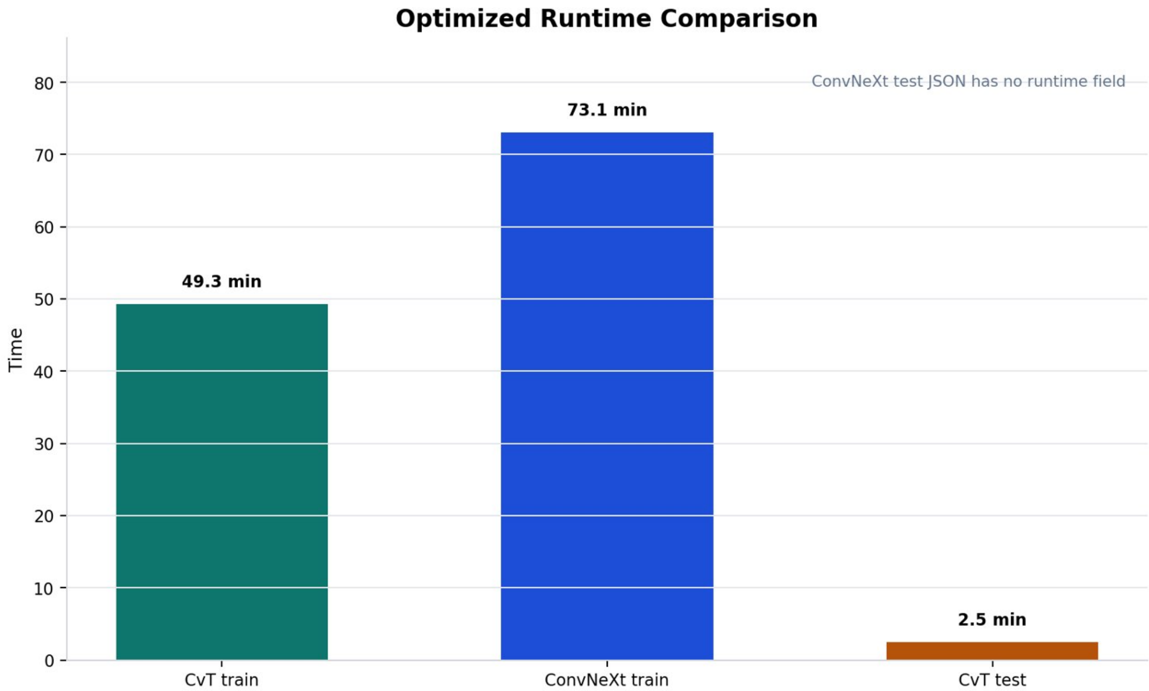

Figure 13: Optimized iNaturalist runtime comparison.

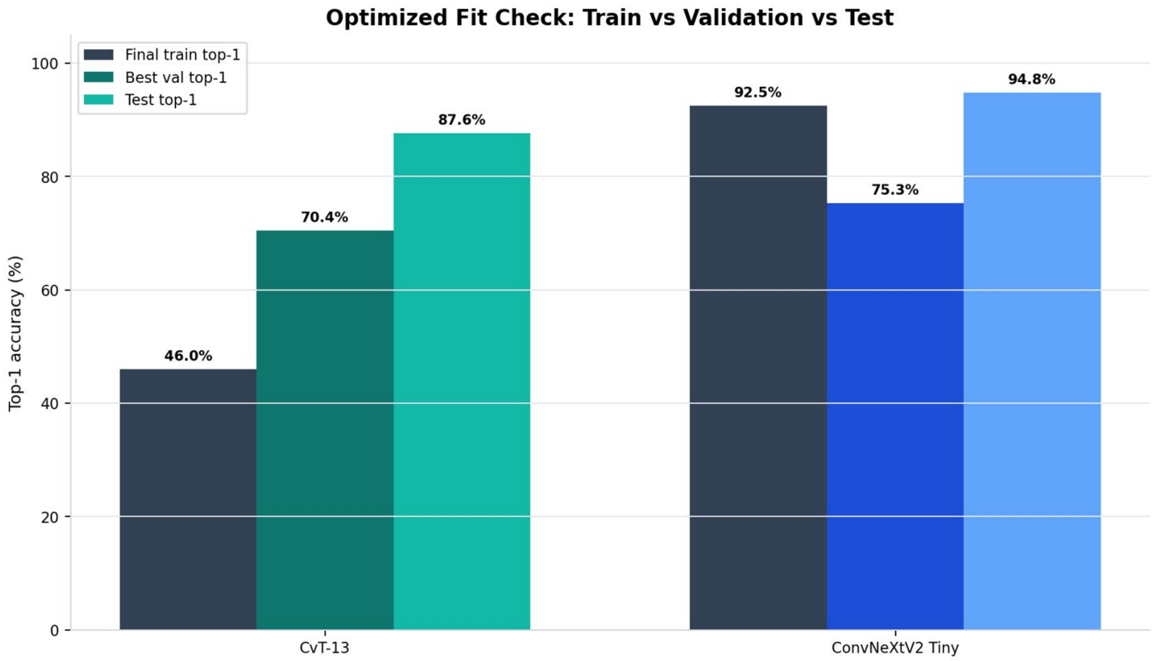

Figure 14: Optimized iNaturalist fit check comparing final train, best validation, and test top-1 accuracy.

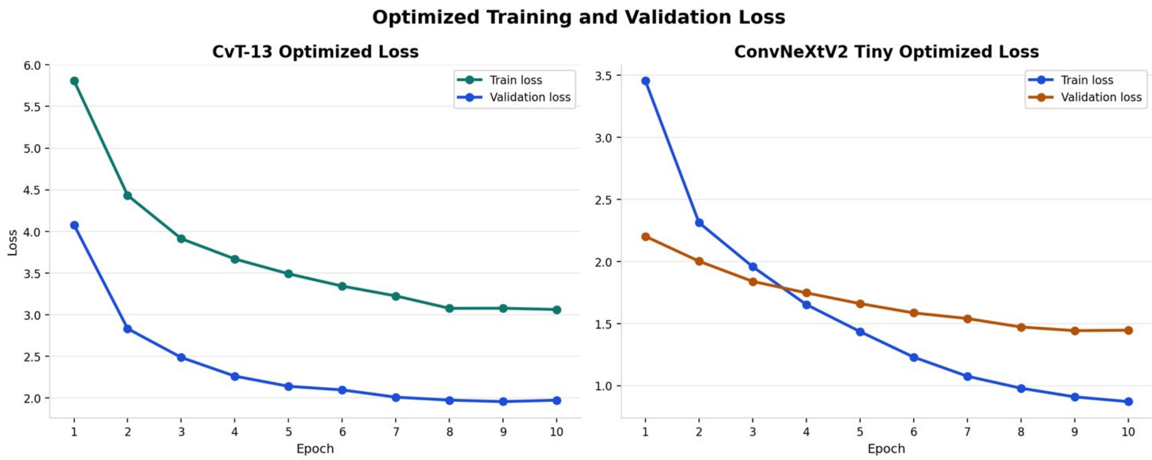

Figure 15: Optimized iNaturalist training loss comparison for both models.

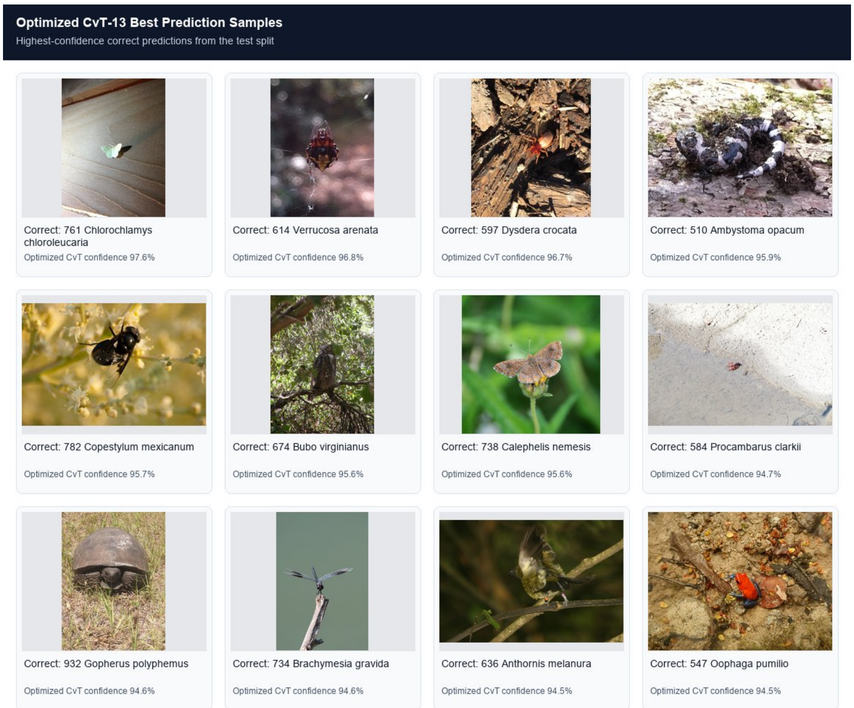

Figure 16: High-confidence correct optimized CvT-13 predictions on the iNaturalist test split.

7 Combined Discussion and Limitations

7.1 What the Current Evidence Already Supports

- the CvT branch preserves complete optimized iNaturalist train/test evidence plus archived Food-101 held-out test evidence,
- the ConvNeXtV2 branch preserves complete optimized iNaturalist evidence and a strong archived Food-101 validation trace,
- the optimized iNaturalist experiment gives a true two-model comparison because both models were trained and tested on the same prepared split,
- the project includes meaningful systems-level optimization rather than only model training.

7.2 Combined Interpretation Across Both Datasets

This final document keeps Food-101 and iNaturalist together in one report, but it presents the saved evidence in separate tables so each result can be interpreted correctly. That structure keeps the report combined at the document level while preventing misleading cross-dataset comparisons inside a single table.

7.3 Main Limitations

Table 9: Main Limitations and Their Effect

Limitation	Why it matters	Handling in this report
Food-101 is retained mainly as archived evidence	The cleaned project no longer preserves the full benchmark bundle for a full rerun.	Report Food-101 conservatively through retained metrics and histories.
Food-101 ConvNeXtV2 lacks matched test JSON	Validation history is not identical to a final held-out test report.	Do not claim a strict Food-101 test ranking for ConvNeXtV2-Tiny.
Dataset difficulty differs	Food-101 and iNaturalist percentages are not the same type of task difficulty.	Compare models within each dataset first.

7.4 Recommended Next Steps

Table 10: Recommended Next Steps

Priority	Action	Expected benefit
High	Restore the separate iNaturalist model code files in the cleaned project folder.	Makes the cleaned repository more complete and easier to reproduce.
High	Run or preserve a matched Food-101 ConvNeXtV2 held-out test evaluation.	Removes the biggest evidence gap in the Food-101 branch.
Medium	Maintain a compact experiment manifest with commands, environment state, and artifact paths.	Improves report regeneration and future reproducibility.

8 GitHub Repository and Team Contributions

8.1 GitHub Repository

GitHub link: https://github.com/Rachith000bharadwaj/DSAI-Deep-Learning

8.2 Team Member Contributions

Table 11: Team Member Contribution Summary

Team Member	USN	Contribution
Rachith Bharadwaj T N	24BDS062	Project architecture, report preparation, and research support.
M G Shree Harsha	24BDS037	Foundation of the project, implementation planning, and comparison design.
Ullas G M	24BDS085	Implemented the model workflows and experiments on the Food-101 dataset.
Shreedhar M Kadkol	24BDS076	Implemented the model workflows and experiments on the iNaturalist dataset.

9 Conclusion

This final report presents Food-101 and iNaturalist together in one structured document and follows a cleaner academic sequence from title page to references. It keeps the stronger figures, restores the GitHub and team contribution page, removes the low-value epoch-wise tables, and presents the saved metrics in a clearer combined-report format.

The strongest complete optimized result remains ConvNeXtV2-Tiny on iNaturalist with 94.840% test top-1 and 98.600% test top-5. CvT-13 remains a strong and stable baseline with 87.600% iNaturalist test top-1 plus archived Food-101 held-out test evidence at 89.624%. The project is therefore strong enough to present as an artifact-backed optimization study across two datasets.

References

[1] L. Bossard, M. Guillaumin, and L. Van Gool, 'Food-101 - Mining Discriminative Components with Random Forests,' ECCV, 2014.

[2] G. Van Horn et al., 'The iNaturalist Species Classification and Detection Dataset,' CVPR, 2018.

[3] H. Wu et al., 'CvT: Introducing Convolutions to Vision Transformers,' ICCV, 2021.

[4] S. Woo et al., 'ConvNeXt V2: Co-designing and Scaling ConvNets with Masked Autoencoders,' CVPR, 2023.

[5] Project repository: https://github.com/Rachith000bharadwaj/DSAI-Deep-Learning